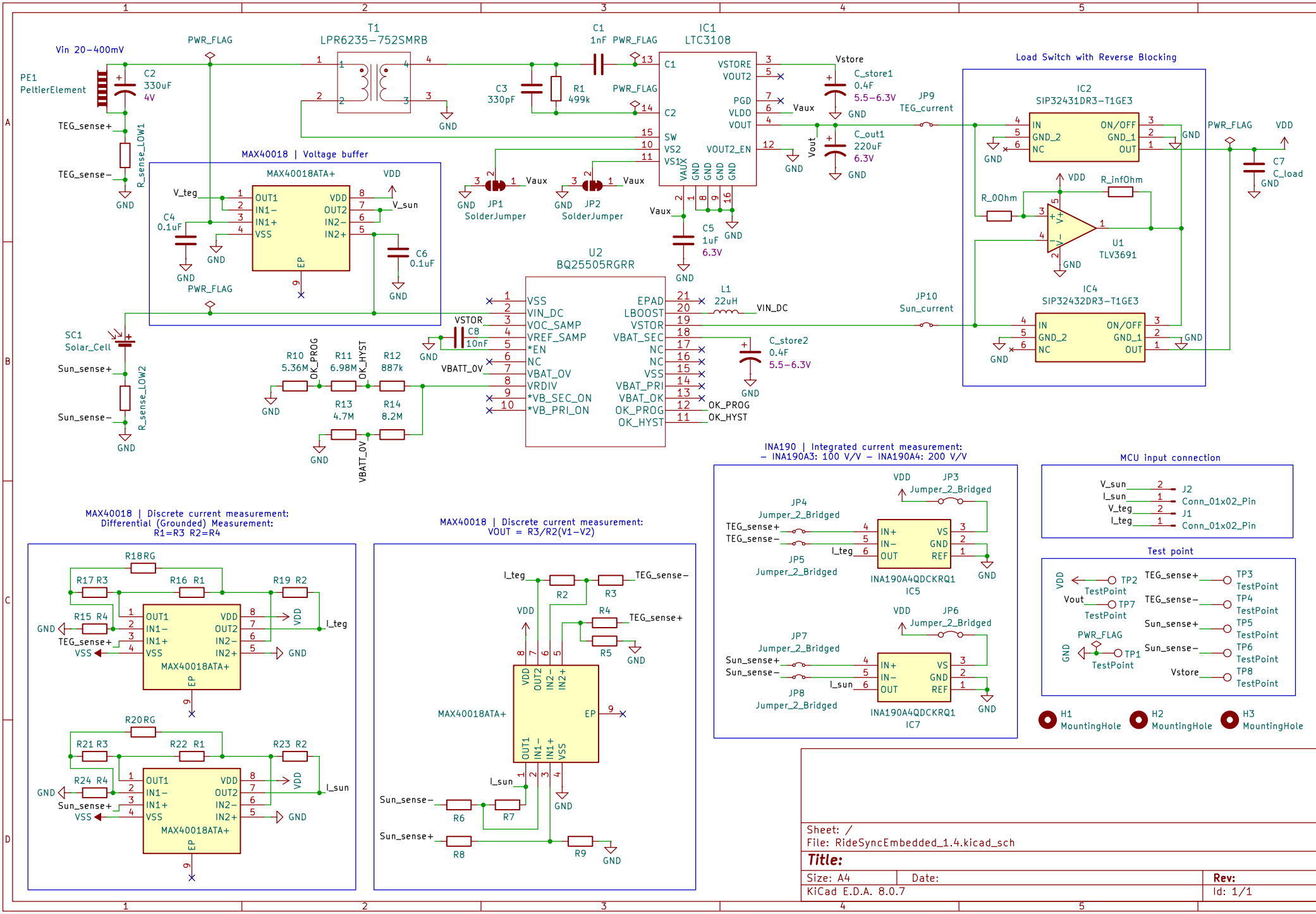
Vin 20–400mV
PE1
PeltierElement
C2
330uF
4V
TEG_sense+
TEG_sense–
R_sense_LOW1
PWR_FLAG
T1
LPR6235–752SMRB
C3
330pF
C1
1nF
R1
499k
IC1
LTC3108
C1
C2
SW
VS2
VS1
VSTORE
VOUT2
PGD
VLDO
VOUT
VOUT2_EN
VAUX
Vstore
C_store1
0.4F
5.5–6.3V
Vaux
C_out1
220uF
6.3V
Vout
JP9
TEG_current
Load Switch with Reverse Blocking
IC2
SIP32431DR3–T1GE3
IN
GND_2
NC
ON/OFF
GND_1
OUT
VDD
R_infOhm
R_0Ohm
U1
TLV3691
IC4
SIP32432DR3–T1GE3
C7
C_load
MAX40018 | Voltage buffer
MAX40018ATA+
V_teg
V_sun
C4
0.1uF
C6
0.1uF
JP1
SolderJumper
JP2
SolderJumper
C5
1uF
6.3V
U2
BQ25505RGRR
L1
22uH
VIN_DC
JP10
Sun_current
SC1
Solar_Cell
Sun_sense+
Sun_sense–
R_sense_LOW2
VSTOR
C8
10nF
VBATT_OV
R10
5.36M
R11
6.98M
R12
887k
R13
4.7M
R14
8.2M
OK_PROG
OK_HYST
C_store2
0.4F
5.5–6.3V
MAX40018 | Discrete current measurement:
Differential (Grounded) Measurement:
R1=R3 R2=R4
R18RG
R17 R3
R16 R1
R19 R2
R15 R4
R20RG
R21 R3
R22 R1
R23 R2
R24 R4
I_teg
I_sun
MAX40018 | Discrete current measurement:
VOUT = R3/R2(V1–V2)
R2
R3
R4
R5
R6
R7
R8
R9
INA190 | Integrated current measurement:
– INA190A3: 100 V/V – INA190A4: 200 V/V
JP3
Jumper_2_Bridged
JP4
Jumper_2_Bridged
JP5
Jumper_2_Bridged
JP6
Jumper_2_Bridged
JP7
Jumper_2_Bridged
JP8
Jumper_2_Bridged
INA190A4QDCKRQ1
IC5
IC7
MCU input connection
J2
Conn_01x02_Pin
J1
Conn_01x02_Pin
Test point
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TestPoint
H1
MountingHole
H2
MountingHole
H3
MountingHole
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Size: A4
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Rev:
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